

Comprehensive 2-Day Workshop Plan

Data Foundations, Analytics, Predictive Modeling & Next-Gen AI Tools

Here is your revised **2-day workshop plan** configured exactly to your schedule requirements: **3 hours per day** (totaling 6 hours). Each day features **2 hours of core curriculum lectures** followed by a **1-hour funny quiz session** designed to keep the students highly engaged.

□ Day 1: Data Analytics (DA) & AI-ML Foundations

□ Session 1 & 2: Data Analytics and AI/ML Core (2 Hours)

This session focuses on data discovery, analytics pipelines, and core machine learning frameworks mapped directly from your presentation resource material.

- **Module 1: Data Analytics (DA) Fundamentals (45 Mins)**
 - **The DA Purpose:** Looking backward at historical datasets to extract insights and analyze specifically what occurred.
 - **Core Tasks:** Cleaning messy data spreadsheets, managing empty or null records, and rendering interactive visual dashboards.
 - **Key Vocabulary:** Defining baseline statistical metrics, identifying recurring directional trends, and staging datasets properly for downstream AI use.
- **Module 2: AI & Machine Learning Foundations (45 Mins)**
 - **What is AI?:** Programming computational systems with capability parameters mirroring organic human decision intelligence.
 - **Everyday Realities:** Analyzing applications built into daily technology such as localized Voice Assistants, facial biometric Face Unlock systems, and digital entertainment Recommendation Engines.
 - **The Big Picture:** Visualizing nested architecture tiers where Deep Learning operates as a localized specialized subset of Machine Learning, which sits broad scale inside the parent Artificial Intelligence block.
- **Module 3: The Three Types of ML & Pipeline (30 Mins)**
 - **The 3 Paradigms:** Supervised Learning (processing labeled target examples), Unsupervised Learning (clustering hidden structural patterns with zero explicit

answers), and Reinforcement Learning (optimizing dynamic behavioral logic based on positive or negative trial-and-error reward signals).

- **The Production Pipeline:** Walking sequential steps chronologically: 1. Collect Data → 2. Prepare Features → 3. Train Model → 4. Test & Evaluate → 5. Predict.
- **Vocabulary Check:** Solidifying exact practical alignment constraints surrounding standard Datasets, computational Features, target output Labels, and the structural threat of model Overfitting.

□ Session 3: The "Brain Break" Funny Quiz (1 Hour)

- **Format:** A highly gamified, interactive competitive trivia block using collaborative response setups like Kahoot! or Mentimeter.
- **Sample Fun Concepts:**
 - **"Identify the AI vs. Reality":** Students race to accurately deduce if an absurd, surreal consumer invention description or pop song lyric was compiled by a broken artificial intelligence neural node or a real human eccentric.
 - **"Overfitting in Real Life":** Reviewing comical cultural scenario examples mapping out situations where an individual hyper-memorizes instructions for one microscopic event constraint but breaks processing functions everywhere else.
 - **"Train a Bad Robot":** A rapid brainstorming game testing student intuition on bad feature architecture—such as calculating a real estate evaluation model fed exclusively on the homeowner's preferred candy bar flavor.

□ Day 2: Data Science (DS) & Modern AI Tools

□ Session 1 & 2: Data Science Frameworks and Next-Gen AI Tools (2 Hours)

This session scales directly up into functional algorithmic engineering, package installation platforms, and modern automated developer utility suites.

- **Module 4: Data Science (DS) & The Popular Models (45 Mins)**
 - **The DS Shift:** Transitioning safely past retrospectively looking at past data logs (DA) toward building sophisticated automated prediction models optimized for looking forward into the future.
 - **Linear Regression:** Plotting directional trends and forecasting a clean continuous numerical vector by fitting a straight target line perfectly through clustered datapoints (e.g., standard housing marketplace validation).
 - **Decision Trees:** Structuring logic pathways into cascading splits of basic yes/no nodes to arrive at complex data inferences (e.g., evaluating auto loan qualifications or diagnostic indicators).
 - **Neural Networks:** Assembling complex calculation matrices loosely modeled after human neurological patterns to confidently map deep non-linear functions (e.g., advanced digital image classification and pattern sorting).
- **Module 5: The Coding Ecosystem (35 Mins)**
 - **The Core Language:** Dissecting the specific mathematical features that make Python the absolute industry standard syntax baseline for modern predictive computing.
 - **Mathematical Array & Frame Management:** Deploying NumPy arrays for massive array matrix mathematics alongside Pandas data structures for organizing clean tabular files smoothly.
 - **Modeling Stack Applications:** Setting up quick statistical frameworks inside the standard Scikit-learn platform, while scaling complex production models directly into robust architectures like TensorFlow and PyTorch.
 - **The Workspace Platform:** Organizing lightweight rapid code prototypes completely online inside zero-installation browser spaces like Google Colab environments.
- **Module 6: Modern AI Tools & Ethical Traps (40 Mins)**
 - **Developer Accelerators:** Leveraging contemporary AI developer assistants (such as ChatGPT pipelines, GitHub Copilot setups, or specialized Cursor environments) to

write high-quality boilerplate software code snippets, eliminate complex algorithmic compilation bugs, and translate undocumented logic structures.

- **Rapid Prototyping Tools:** Using easy visual tools such as Google's Teachable Machine UI to build robust standalone camera and microphone object classifiers without manual script overhead.
- **Responsible Engineering:** Addressing systemic dataset bias patterns, establishing high criteria for baseline algorithm fairness, and securing corporate user data privacy controls from leakage vectors.

□ **Session 3: The "Future Tech" Funny Quiz (1 Hour)**

- **Format:** A high-energy team challenge designed explicitly to explore real development hurdles and weird logic flaws in upcoming technologies.
- **Sample Fun Concepts:**
 - **"Spot the Copilot Fail":** Presenting teams with hilariously over-complicated, logic-defying, or completely broken blocks of script generated by bad prompts, where teams must sprint to find the hidden structural glitch.
 - **"The AI Ethicist Dilemma":** Presenting teams with highly exaggerated ethical choices (e.g., 'Your autonomous test vehicle navigation system must instantly optimize between running directly into a wagon load of fresh golden retriever puppies or a heavy cart filled with artisanal triple-layer chocolate cakes—what priority score does the algorithm compile?').
 - **"Explain Like I'm 5":** Teams compete head-to-head to break down highly technical structural configurations like multi-layered neural networks or pipeline data frames using nothing but everyday kitchen items or classic cartoons as literal analogies.